

Institute/Department	School of Computer Science Engineering	Program	Bachelor of Technology (Computer Science and Engineering-Artificial Intelligence and Machine Learning) with Microsoft(LCS202)
Master Subject Coordinator Name:	Ghufran Khan	Master Subject Coordinator E-Code:	T550003
Course Name	Cloud Security	Course Code	25CST-206

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
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Course Type	Course Category	Mode of Assessment	Mode of Delivery
Minor Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1. Develop and deliver comprehensive curricula that equip students with the knowledge and skills necessary for successful careers in computer science and engineering areas. M2. Implement innovative teaching methodologies to foster critical thinking and problem-solving abilities among students. M3. Encourage faculty and students to engage in pioneering research projects that contribute to technological advancements. M4. Facilitate partnerships with other departments and institutions to conduct convergent research that bridges multiple disciplines. M5. Organize seminars, workshops, and collaborative projects that encourage cross-disciplinary engagement. M6. Establish platforms for alumni to actively participate in departmental activities, such as mentoring programs and guest lectures. M7. Maintain regular communication with alumni to foster a supportive and engaged community.
Vision of the Department	V1. Provides undergraduate and graduate students with a high-quality education leading to rewarding, high-paying careers. V2. Engages faculty in cutting-edge research and innovation, actively involving and training students in these endeavors. V3. Serves as a hub where convergent research themes intersect, enabling advancements across various disciplines. V4. Maintains active engagement with alumni, fostering a strong and supportive community

Program Educational Objectives(PEOs)

PEO1	Graduates of Computer Science and Engineering will contribute to the Nation's growth through their ability to solve diverse and complex computer science and engineering problems across a broad range of application areas.
PEO2	Graduates of Computer Science and Engineering will be successful professionals, designing and implementing Products & Services of global standards in the field of Computer Science & Engineering, becoming entrepreneurs, Pursuing higher studies & research.
PEO3	Graduates of Computer Science and Engineering Program will be able to adapt to changing scenarios of dynamic technology with an ability to solve larger societal problems using logical and flexible approach in decision making.

Program Specific OutComes(PSOs)

PSO1	Apply core concepts of computer science, including programming, algorithms, database systems, and software engineering, to design and develop efficient and innovative computational solutions.
PSO2	Integrate Artificial Intelligence and Machine Learning techniques into software and system designs to enhance functionality, enable intelligent automation, and solve complex real- world challenges

Program OutComes(POs)

PO1	Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problem
PO2	Ability to identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PO4	Ability to conduct investigation into complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Ability to create, select and apply appropriate techniques, resources and modern engineering, and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.
PO6	Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Ability to recognize and incorporate the diversity and commonalities of engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO8	Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	Ability to function effectively as an individual and as member or leader in diverse teams and interdisciplinary settings.
PO10	Ability to communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Ability to demonstrate knowledge and understanding of the engineering and management principles and also apply these principles to one's own work as member and leader to the teams to manage projects and interdisciplinary teams.
PO12	Ability to recognize the need for, and have the preparations and ability to engage in independent and lifelong learning in the broadest context of technological change.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance	Tim Mather, Subra Kumaraswamy, Shahed Latif	1	O'Reilly Media	2009
2	Cloud Security Alliance CSA Cloud Controls Matrix (CCM) Cloud Security Alliance	Thomas Sager	8.1	CIS	2011

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Cloud Security For Dummies	Ted Coombs	1	For Dummies	2022

Course OutCome	
SrNo	OutCome
CO1	CO1
CO2	CO2
CO3	CO3
CO4	CO4
CO5	CO5

Lecture Plan Preview-Theory						
Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	UNIT 1: Evolution and Definition of Cloud Computing,	Introduction to Cloud Computing	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO1

1	2	UNIT 1: Evolution and Definition of Cloud Computing,	Evolution of Cloud Computing	,T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An,R-Cloud Security For Dummies	PPT	CO1
1	3	UNIT 1: Evolution and Definition of Cloud Computing,	Cloud Characteristics	,T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An,R-Cloud Security For Dummies	PPT	CO1
1	4	UNIT 1: Evolution and Definition of Cloud Computing,	Cloud Service Models	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO1
1	5	UNIT 1: Evolution and Definition of Cloud Computing,	Cloud Deployment Models	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO1
1	6	UNIT 1: Evolution and Definition of Cloud Computing,	Major Cloud Providers	R-Cloud Security For Dummies	PPT	CO1
1	7	UNIT 1: Evolution and Definition of Cloud Computing,	Shared Responsibility Model	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO1
1	8	UNIT 1: Evolution and Definition of Cloud Computing,	Security Challenges in Cloud	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO1
2	9	UNIT 2: Identity and Access Management (IAM)	IAM Principles	R-Cloud Security For Dummies	PPT	CO2
2	10	UNIT 2: Identity and Access Management (IAM)	IAM Architecture	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	11	UNIT 2: Identity and Access Management (IAM)	Authentication Mechanisms	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	12	UNIT 2: Identity and Access Management (IAM)	MFA	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	13	UNIT 2: Identity and Access Management (IAM)	Identity Federation	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	14	UNIT 2: Identity and Access Management (IAM)	Directory Services	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	15	UNIT 2: Identity and Access Management (IAM)	PAM	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2

2	16	UNIT 2: Identity and Access Management (IAM)	Encryption Basics	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
2	17	UNIT 2: Identity and Access Management (IAM)	Key Management Concepts	,T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An,R-Cloud Security For Dummies	PPT	CO2
2	18	UNIT 2: Identity and Access Management (IAM)	Unit Review/Case Study	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO2
3	19	UNIT 3: Virtual Private Cloud (VPC)	VPC Architecture	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
3	20	UNIT 3: Virtual Private Cloud (VPC)	Network Segmentation	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
3	21	UNIT 3: Virtual Private Cloud (VPC)	Network Isolation	,T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An,R-Cloud Security For Dummies	Case Study,PPT	CO3
3	22	UNIT 3: Virtual Private Cloud (VPC)	Security Groups	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
3	23	UNIT 3: Virtual Private Cloud (VPC)	Network ACLs	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	Infographics, PPT	CO3
3	24	UNIT 3: Virtual Private Cloud (VPC)	Cloud Firewalls	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT,Reports	CO3
3	25	UNIT 3: Virtual Private Cloud (VPC)	IDS/IPS	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
3	26	UNIT 3: Virtual Private Cloud (VPC)	VPN and Direct Connect	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
3	27	UNIT 3: Virtual Private Cloud (VPC)	DDoS Protection & Mitigation	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO3
4	28	UNIT 4: Data Classification and Labeling,	Data Classification	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	29	UNIT 4: Data Classification and Labeling,	Data Labeling	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	30	UNIT 4: Data Classification and Labeling,	Data Loss Prevention	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	31	UNIT 4: Data Classification and Labeling,	Database Security	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	32	UNIT 3: Virtual Private Cloud (VPC)	Secure Data Storage	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	33	UNIT 4: Data Classification and Labeling,	Archiving	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	34	UNIT 4: Data Classification and Labeling,	Logging Architecture	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	35	UNIT 3: Virtual Private Cloud (VPC)	Cloud Native Logging	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4

4	36	UNIT 4: Data Classification and Labeling,	SIEM	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO4
4	37	UNIT 4: Data Classification and Labeling,	Threat Detection, Metrics & Reporting	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	Case Study,Infogr aphics,PPT	CO4
5	38	UNIT 5: Security by Design Principles	Security by Design	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	39	UNIT 5: Security by Design Principles	Zero Trust Model	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	40	UNIT 5: Security by Design Principles	Application Security	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	41	UNIT 5: Security by Design Principles	DevSecOps Integration	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	42	UNIT 5: Security by Design Principles	Continuous Security Monitoring	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	43	UNIT 5: Security by Design Principles	Vulnerability & Patch Management	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	44	UNIT 5: Security by Design Principles	Security Automation & CSPM	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5
5	45	UNIT 5: Security by Design Principles	Emerging Trends in Cloud Security	T-Cloud Security Alliance CSA CI,T-Cloud Security and Privacy: An	PPT	CO5

Sr No	Assessment Name	Exam Name	Max Marks
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CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	NA	2	2	1	1	NA	1	NA	2	2	1
CO2	3	2	3	2	3	2	NA	2	NA	1	NA	2	3	3
CO3	3	3	3	2	3	2	1	1	2	2	2	2	2	3
CO4	2	3	2	3	3	2	1	1	2	1	1	2	3	3
CO5	3	3	3	3	3	2	2	2	2	2	2	3	3	3
Target	2.6	2.4	2.4	2.5	2.8	2	1.25	1.4	2	1.4	1.67	2.2	2.6	2.6

